Optics & Laser Technology 177 (2024) 111063

Contents lists available at ScienceDirect

Optics and Laser Technology

journal homepage: www.elsevier.com/locate/optlastec

Full length article

Femtosecond laser welding of sapphire-copper using a thin film titanium interlayer

Han Yu, Jia-Xuan Zhao, Lin-Jie Zhang *, Suck-Joo Na

State Key Laboratory for Mechanical Behavior of Materials, Xi'an Jiaotong University, Xi'an 710049, China

ARTICLE INFO

Keywords:
Sapphire
Copper
Femtosecond laser-induced micro-welding
Laser-induced backward transfer
Thin film Titanium interlayer

ABSTRACT

The ability to weld metals and sapphire is widely required in aerospace, micro-electro-mechanical systems and microwave energy delivery devices. The experiments of femtosecond laser welding between sapphire and copper (Cu) were conducted under non-optical contact conditions. And the effect of pre-fabricating a Ti film on the sapphire surface using femtosecond laser on the bonding strength of ssapphire/Cu joint was investigated. The interface of the sapphire/Cu directly welded by femtosecond laser contains a mixture of sapphire and copper, and the shear strength of the sapphire/Cu interface was approximately 46.4 MPa. The results indicated that the laser-induced molten state of sapphire and the laser-ablated copper mixed together at the interface, filling the gaps and forming an effective metallurgical bond. And then, in the case that the Ti film was prepared on the sapphire surface through femtosecond laser-induced backward transfer (LIBT), a Ti transition layer between sapphire and cooper was observed on the bonding interface of the joint, and the shear strength of the joint increased to approximately 115.5 MPa. A mixture of sapphire, copper, titanium, and titanium oxide was observed at the interface of the joint. The Ti layer between sapphire and copper may effectively alleviated residual stresses at the interface and promoted bonding between copper and sapphire, thereby improving the strength of the joint. The methods and results of this study have reference value for the ultrafast laser bonding between other transparent brittle materials and metals.

1. Introduction

Sapphire, which is of high light transmittance, high strength, high hardness, corrosion resistance, and high-temperature resistance, has been widely applied as the material to produce optical windows in radio radars, photoelectric instruments, high-precision instruments, smart camera lenses, and microwave energy delivery windows [1–3]. As a material for producing optical windows, sapphire generally needs to be reliably welded with metals, so as to realize the expected structures and functions [4]. For example, the microwave energy delivery window in space traveling-wave tubes (TWTs) is formed by welding sapphire and a copper (Cu) cavity together. The welding quality of the microwave energy delivery window matters in the service life, output power, and stability of space TWTs [3].

Sapphire and Cu differ greatly in the type of chemical bond and the coefficient of thermal expansion, (sapphire ($6.6 \times 10^{-6}K^{-1}$), Cu ($17.5 \times 10^{-6}K^{-1}$)) it is difficult to achieve welding of the two. At present, diffusion welding and brazing are the most common approaches to weld sapphire and Cu [5–7]. In the two approaches, structures need to be heated as a whole in an atmosphere-controlled furnace, followed by heat preservation for a period of time under pressure and then cooling. The machining process is not only complex and inefficient, but also enlarges the broadband dielectric loss of the energy input window of sapphire [3]. In the meantime, brazed and diffusion-welded joints generally face problems including the poor high-temperature properties and large residual stress gradient on the bonding interface [8].

Femtosecond laser micromachining, characterized by advantages including a small heat effect and high peak power density, can induce the nonlinear absorption effect and local phase change of transparent materials and is an ideal method to achieve micro-welding of transparent materials [9,19]. Takayuki *et al.* (2005) [9] in Osaka University (Japan) were the first to propose and demonstrate the application of ultrashort-pulse laser to micro-welding of transparent glass materials and they fused glass in local areas by focusing the laser onto the interface between two substrates, thus realizing direct welding of transparent glass materials. In 2008, Germany scholar Alexander *et al.* [10,11] promoted the ultrashort-pulse laser-induced micro-welding to the welding of glass and monocrystalline silicon. And in 2023, Pan *et al.* [21]

* Corresponding author.
E-mail address: zhanglinjie@mail.xjtu.edu.cn (L.-J. Zhang).

https://doi.org/10.1016/j.optlastec.2024.111063
Received 23 October 2023; Received in revised form 18 March 2024; Accepted 22 April 2024
Available online 26 April 2024

achieved the direct connection between sapphire and Fe-36Ni alloy for the first time using femtosecond laser micro-welding technology. Optical contact (with the gap smaller than λ/4) is acknowledged as an important factor that affects the quality of ultrashort-pulse laser-induced welding. Under optical contact, surface ablation when applying ultrashort-pulse laser onto the surfaces of the two materials can be effectively inhibited, thus improving the welding quality. However, optical contact sets a high requirement for surface evenness and roughness of samples, which hinders the industrial application and popularization of the technology. Soren *et al.* [12] in University of Jena (Germany), Chen *et al.* [13] in Irey Watt University (UK), and Duan *et al.* [14] in Huazhong University of Science and Technology (China) successively used the ultrashort pulse laser at high repetition frequency to realize the micro-welding of transparent glass materials under conditions with no need of optical contact. Li *et al.* [15] adopt a multi-scan picosecond laser welding method to weld the non-optical soda lime glass that has a gap up to ~5 μm. Shen *et al.* [16] proposed a method that combined laser welding with ultrafast laser welding to achieve a connection between glass and metal with a gap in the range of hundreds of micrometers. Wu *et al.* [17] placed a slit near the front of the objective lens to shape the incident circular Gaussian light into an elliptical Gaussian distribution, which could effectively inhibit the induction and generation of microcracks. Zhang *et al.* [18] proposed that mJ-level pulsed femtosecond laser transmission welding can enhance joint strength, eliminate the need for special surface preparations, and potentially overcome the challenge of aluminum-fused silica welding with a large interface gap. Wang *et al.* [20] not only reduce the fracture of fused silica and increase shear strength up to 25.75 MPa by means of focal point pre-compensation, but also demonstrate the extraordinary potential of ultrashort laser welding in the field of precise connection of heterogenous materials.

Furthermore, it is necessary to explore methods to improve the interface bonding strength, reduce residual stress, and optimize the interface formation. Based on this, deeper insights into the physical mechanisms of connecting transparent materials and metal materials through femtosecond laser under optical contact conditions need to be gained. Adding an intermediate transition layer is an effective method to reduce the residual stress on dissimilar joints and the stress gradient on interfaces. Liu *et al.* [22] added a niobium (Nb) interlayer in the diffusion-welded joint of Al_2O_3 ceramics and Cu, which improves the bond strength and plasticity of Cu/Al_2O_3 joints. In the industry, the common methods to produce a metalized layer on glass including sapphire mainly include electroplating and chemical plating. Traditionally, these methods need lots of steps, including wetting, sensitization, photooxidation, activization, and metallization. The ultrashort-pulse laser shows its unique advantages in the development of non-contact surface graphic technology. Laser-induced backward transfer (LIBT) refers to irradiating a laser beam onto the donor and target after passing through a transparent acceptor material. After being ablated, the donor and target surfaces are transferred onto the transparent material surface as small liquid droplets or vapors, thus rapidly preparing metallized coating on the transparent acceptor material surface. Rodriguez *et al.* [23] formed conductive and elastic glass-graphene nanocomposites by using LIBT. Therefore, a method that using femtosecond LIBT to prefabricate metallized slow-release coating on the sapphire surface was adopted to assist the ultrafast laser-induced micro-welding of sapphire and Cu.

The research attempted to use femtosecond laser to weld sapphire and Cu, and analyzed the formation mechanism of the bonding interface of the transparent material and metal under the non-optical contact conditions. Meanwhile, to improve the bonding strength of joints, the method of prefabricating micro-textures on the metal surface and prefabricating metallized slow-release coating on the sapphire surface was adopted to assist femtosecond laser-induced micro-welding of sapphire and Cu. The research results provide technological guidance and theoretical support for ultrafast laser-induced micro-welding of sapphire and Cu and also provide reference for the welding of other transparent sapphire and metals.

2. Materials and methods

2.1. Test materials

Sapphire used in the tests was C-face α-Al_2O_3 monocrystalline sapphire, with the purity not smaller than 99.99 % and dimensions of 10 mm × 10 mm × 0.5 mm. After twin polishing, the surface roughness Sa of sapphire is lower than 10 nm. Before femtosecond laser-induced micro-welding and femtosecond LIBT tests of coating, sapphire was ultrasonically cleaned with acetone for 30 min to remove oil contamination and then dried. Cu used in the tests was T2 Cu with the purity not lower than 99.99 %. After fine grinding, polishing, ultrasonic cleaning with acetone, and drying before tests, the surface roughness S_a of Cu was 150 nm.

2.2. Test devices and methods

Fig. 1(a) shows the surface micro-textures prepared by femtosecond laser micromachining, the coating prepared by femtosecond LIBT, and the femtosecond laser micromachining system used for femtosecond laser-induced micro-welding. The wavelength and pulse width of the femtosecond laser are separately 1,030 nm and 400 fs. The optical system during femtosecond laser-induced micro-welding is displayed in Fig. 1(b). For the purpose of comparative research, three types of joints were prepared. Table 1 shows the preparation process and technological parameters of the three types of joints.

2.2.1. Femtosecond laser-induced micro-welding: Joint 1#

Before welding, the transparent sapphire was superimposed on Cu. The sapphire was not assembled using a fixture and no load was applied on the samples. In the welding process, after the femtosecond laser beam firstly passed through a triple beam expander and then a reflector, the laser spot changed to propagate vertically downwards. After passing through a focusing lens with the focal length of 100 mm, the light spot was focused on the sapphire/Cu interface, with the diameter of 20 μm. The energy density of the light spot followed Gaussian distribution. During femtosecond laser-induced micro-welding, the light spot reciprocated along a serpentine pattern, with the spacing of 100 μm between adjacent weld passes. The total welding area was a square area measuring 7 mm × 7 mm. The entire welding tests were conducted in the air environment. The welding schemes of Joint 2# was completely same as that of Joint 1#.

2.2.2. Preparing Ti coating using femtosecond LIBT and laser welding: Joint 2#

It can be seen from Fig. 2(b) that the pure Ti sample was put on the working platform and then the sapphire sample was placed above the Ti sample. The upper surface of the Ti sample has a spacing of 0.15 mm with the lower sample of the sapphire sample. The femtosecond laser was incident onto the Ti sample, with the defocus of −4 mm. In the femtosecond LIBT process, the light spot reciprocated along a serpentine pattern and the spacing between adjacent weld passes was 100 μm. The total scanning area was a square area measuring 7 mm × 7 mm. Afterwards, the Cu sample was put on the working platform and then sapphire was superimposed on the Cu surface. The Ti-coated sapphire surface was placed downwards and closely fitted with the upper surface of Cu. Finally, sapphire and Cu were welded using the scheme completely same as that of Joint 1#.

2.2.3. Analysis methods of femtosecond laser-induced micro-welded joints

After finishing the welding tests, the diamond wire cutting method was adopted to cut cross sections along the direction vertical to the welding direction at the joint. After polishing, a SU8230 scanning

(a)

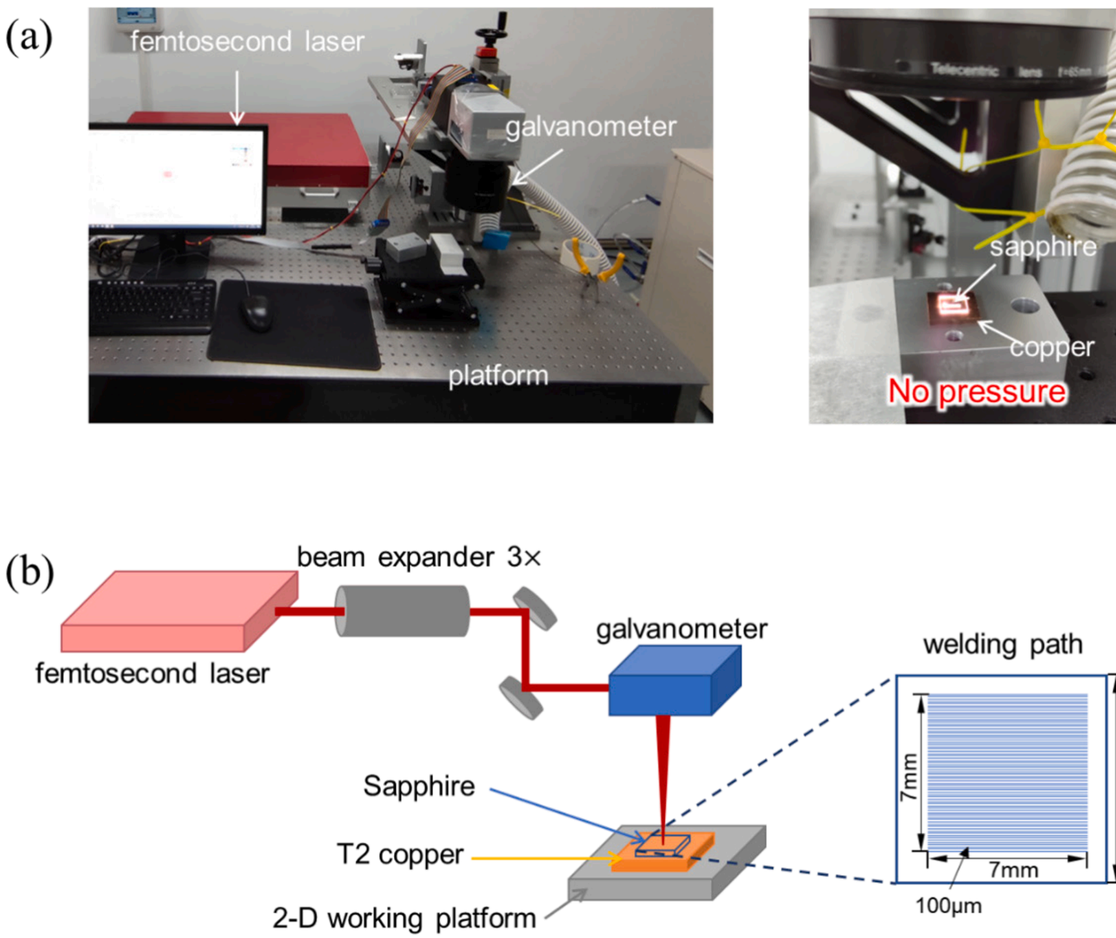


Fig. 1. Schematic diagrams for (a) the test scenario of femtosecond laser micromachining and (b) femtosecond laser-induced micro-welding process.

Table 1
Parameters during femtosecond laser surface micro-texturing, surface coating, and welding.

Steps	Parameters	Joints	
		Joint 1#	Joint 2#
Pre-treatments	Pre-treatment processes	/	LIBT
	Power/W	/	2.5
	Scanning speed v/mm/s	/	10
	Pulse width/fs	/	400
	Pulse frequency/kHz	/	250
	Defocus/mm	/	−4
	Wavelength/nm	/	1030
	Scanning spacing/μm	/	100
	Gap between glass and target/mm	/	0.15
	Path pattern	/	Serpentine
	Atmosphere	/	Air
Welding	Power/W	15	15
	Scanning speed v/mm/s	5	5
	Pulse width/	400	400
	Pulse frequency/kHz	500	500
	Defocus/mm	0	0
	Wavelength/nm	1030	1030
	Scanning spacing/μm	100	100
	Path pattern	Serpentine	Serpentine
	Clamping	Free	Free
	Atmosphere	Air	Air
Shear strength / MPa		46.4	115.5

electron microscope equipped with an energy dispersive spectrometer (EDS) was used to observe cross sections and shear fractures of joints. Then, mechanical properties of the joints were tested and the shear strength of joints was tested using the cutting method. The dimensions of samples after being cut using the diamond wire cutting method are shown in Fig. 3(a). The shear samples were placed on an INSTRON MODEL 1186 universal testing machine to carry out compression-shear tests, during which the clamping mode is illustrated in Fig. 3(b). In the tests, the loading rate was 0.5 mm/min and the shear strength was calculated using $\tau = \frac{F}{S} = \frac{F}{LW}$, where τ, F, S, L, and W separately represent the shear strength (MPa), fracturing load (N), welding area (mm^2), welding length (mm), and welding width (mm). To study structural changes of sapphire in the bonding area, a laser Raman spectrometer was adopted to record Raman spectra in the sapphire side in the spectral region with the laser wavelength of 532 nm and room temperature of 300 cm^{-1}–800 cm^{-1}.

3. Results

Parameters in Table 1 were used to prepare three types of micro-welded sapphire/Cu joints. Fig. 4 shows macro-morphologies of the two types of joints. It can be seen that the sapphire surfaces in the two types of joints do not have macro-cracks and the weld zones are off-white, which is totally different from the original surface color of Cu samples.

3.1. Direct femtosecond laser-induced micro-welding

Fig. 5 shows the full view of the cross section of direct femtosecond

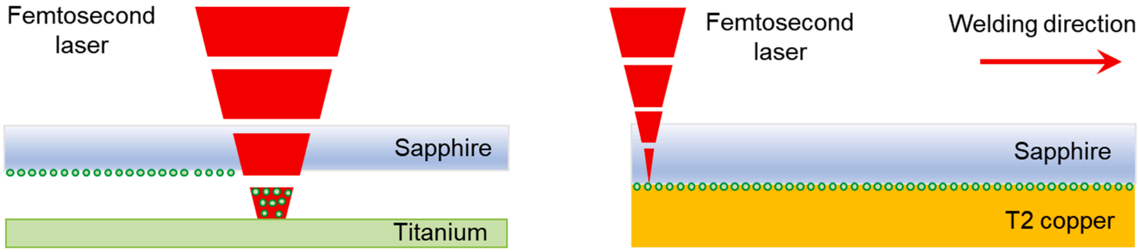


Fig. 2. Schematic diagrams for welding after femtosecond LIBT of coating.

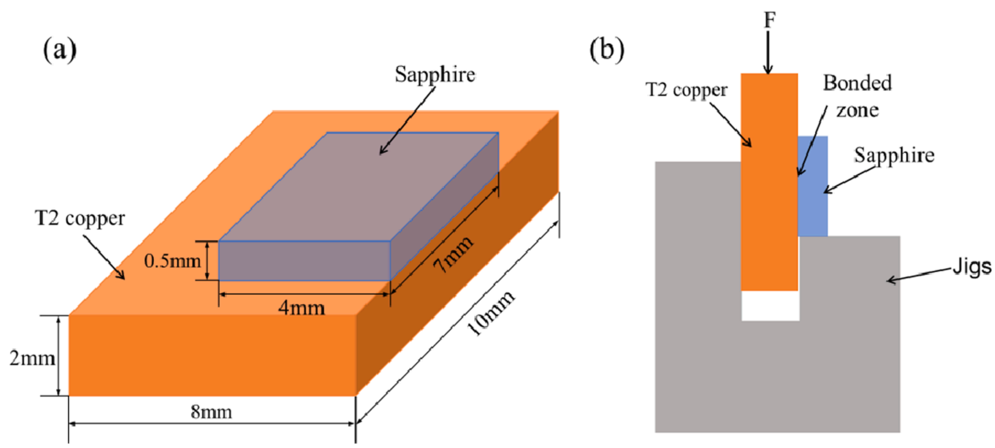


Fig. 3. Sample shape and size, and test scenario of compression-shear tests.

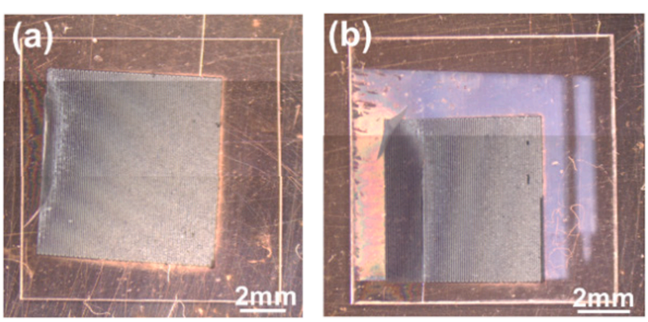


Fig. 4. Surface macro-morphologies of three types of femtosecond laser-induced micro-welded sapphire/Cu joints. (a) Joint 1#: femtosecond laser-induced direct micro-welded joint; (b) Joint: 2#: femtosecond laser-induced micro-welded joint of Ti-coated sapphire and Cu.

laser-induced micro-welded sapphire/Cu joint (Joint 1#) under non-optical contact conditions. Fig. 6 further displays the micro-morphologies at high magnification near the sapphire/Cu interface on the cross section of Joint 1#.

It can be seen from Fig. 5(b) that ① periodic welded zones in irregular shapes are found in the sapphire side near the sapphire/Cu interface on the cross section and the spacing between two adjacent welded zones is 100 μm, which is the spacing between adjacent scanning paths in the welding process. ② It can be seen from Fig. 5(c) that each welded zone in the direction parallel to the interface is about 80 μm wide, which is much larger than the diameter of 20 μm of the focus of femtosecond laser. The height of welded zones in the direction vertical to the interface generally does not exceed 100 μm. This means that the welded zones are near the interface while do not extend inside sapphire. ③ It can be seen from Fig. 5(f) that a few island-like Cu-rich areas are dispersed in welded zones, which suggests that Cu has entered the welded zones. Further observation of the location of island-like Cu-rich areas in the right upper part of Fig. 5(f) reveals that Cu has entered far from the sapphire/Cu interface. ④ As shown in Fig. 6(b) and (c), the surface morphologies of sapphire near the sapphire/Cu interface (top of Fig. 6(b)) exhibit typical morphologies formed after melting and solidification.

Fig. 6 further displays micro-morphologies at high magnification near the sapphire/Cu interface on the cross section of Joint 1#. As displayed in Fig. 5(b), black and gray matrices are present at the sapphire/Cu interface, in which the black matrix contains lots of gray island-like areas with the size of 0.5 ~ 2 μm. EDS point scanning results show that the black and gray matrices at the interface are separately Al_2O_3 and Cu, while the gray island-like areas in the black matrix Al_2O_3 are Cu. The composition distribution in Fig. 6(c) and line scanning results at the interface in Fig. 6(d) show that favorable metallurgical bonding is formed at the interface. Phase compositions at the interface were studied through micro-Raman spectroscopy. As shown in Fig. 6(e), Raman spectral peaks at 378, 417, 643, and 749 cm^{-1} in the sapphire side at the interface are all spectral peaks of sapphire while other phases are not formed.

3.2. Joint 2# with Ti-coated sapphire surface

It can be seen from Fig. 7(b) that Ti coating is successfully plated on the sapphire surface through LIBT. The morphologies of Ti-coated sapphire are shown in Fig. 7(a). The light blue area on sapphire is the metallic Ti coating. As displayed in Fig. 7(b) and (c), Ti on sapphire surface is nano-sized spherical particles with the diameter of 20 nm–100 nm. They are uniformly and densely distributed on the sapphire surface. According to the statistics, the density of Ti particles on sapphire is about 98 NPs/μm^2.

Fig. 8 displays the cross-sectional morphologies of Joint 2# of Ti-coated sapphire and Cu prepared through femtosecond laser-induced micro-welding. It can be seen that Ti-coated sapphire and Cu are successfully welded. As shown in Fig. 8(b), the gap at the interface reduces obviously, so do the damage zones in the sapphire side. It can be seen

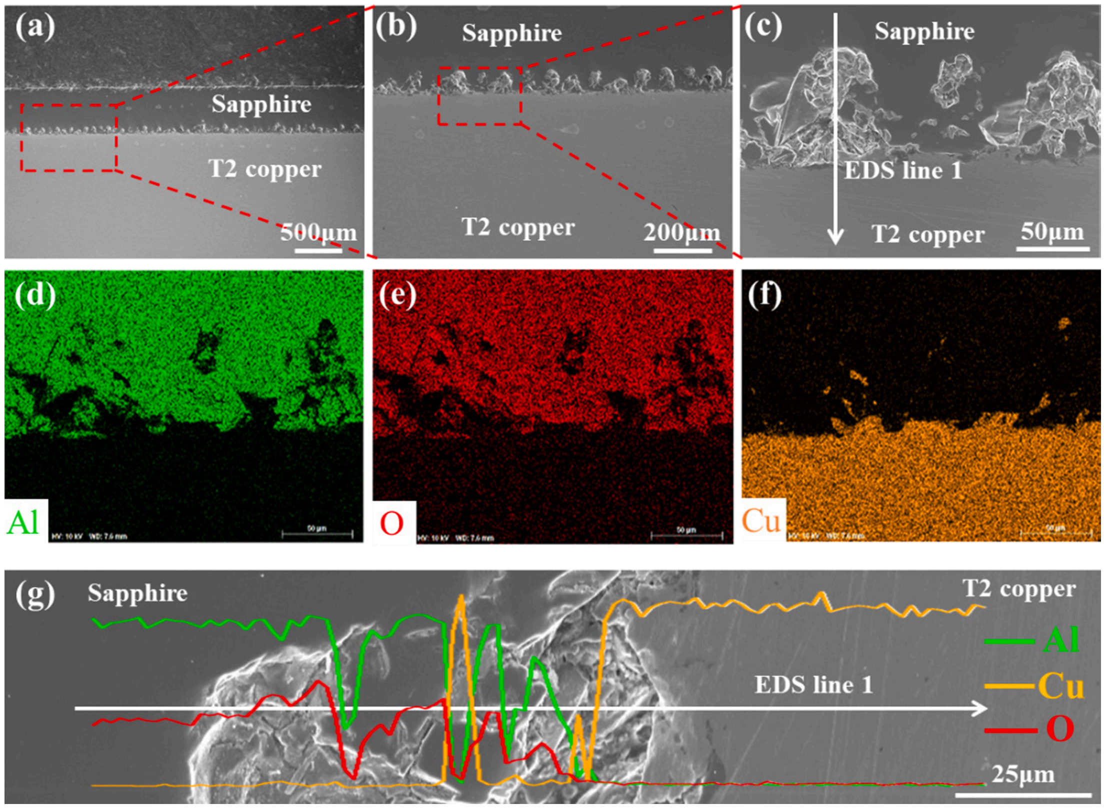


Fig. 5. (a)–(c) Micro-morphologies in the bonding area, (d)–(f) composition distribution in the bonding area, and (g) EDS line scanning results of the joint prepared by direct femtosecond laser-induced micro-welding (Joint 1#).

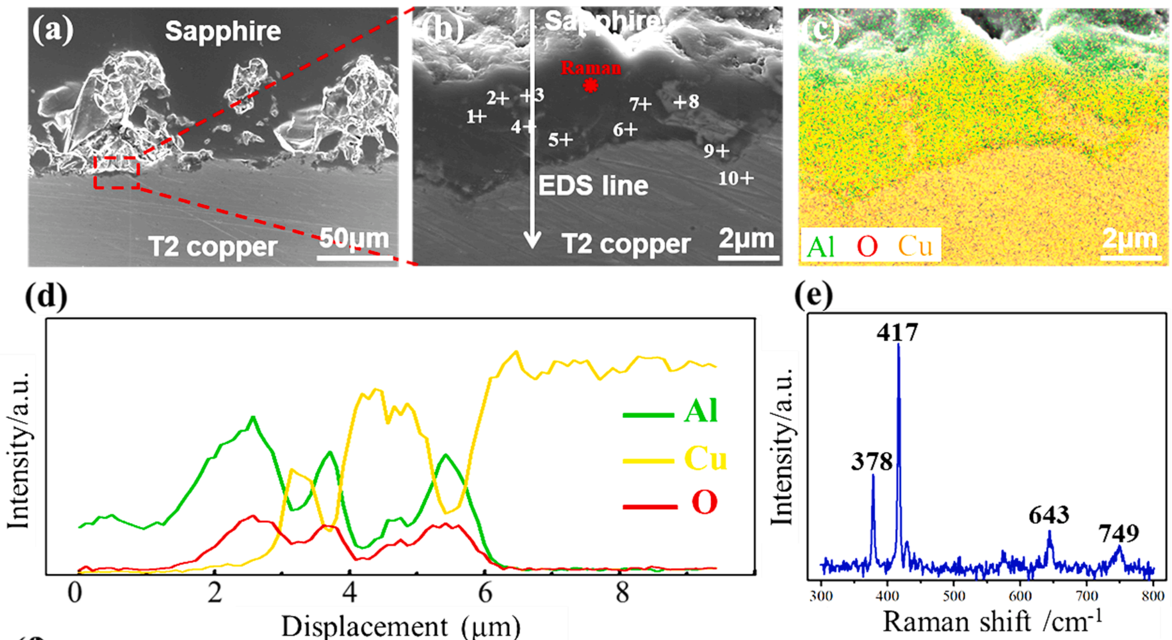
that Cu at the interface is protuberant upwards. Different from Joint 1# prepared through direct femtosecond laser-induced micro-welding, Cu particles are not observed to enter sapphire on the cross section of Joint 2#. This is probably because due to the presence of the Ti coating on the sapphire surface, the laser transmittance decreases and the energy acted on the sapphire surface declines after the laser passes through sapphire and Ti coating. As a result, the number of Cu particles splashed upwards reduces and the upward impact of plasmas is weakened. Fig. 8(i) displays the magnified micro-morphologies at the interface of Joint 2#. It can be seen from composition distribution that the sapphire/Cu interface contains Ti coating of 0.5 μm thick, which suggests that Ti is not fused in sapphire but exists independently in the femtosecond laser-induced micro-welding process. In the thermal cycle process, the coefficient of thermal expansion of Ti is somewhat between those of sapphire and Cu. The layered Ti can well relieve the heat effect. Meanwhile, the element distribution results also indicate that in the femtosecond laser-induced micro-welding process, some Ti diffuses in sapphire. When the temperature near the joint exceeds the melting point of sapphire, the Ti coating diffuses into sapphire. At a high temperature, Ti can form a TiO-TiO_2 composite reaction layer at its interface with sapphire, thus improving the bonding strength of the interface.

4. Discussion

The compressive and shear properties of the two types of joints were discussed. The load–displacement curves of the three types of joints are illustrated in Fig. 9. The shear strength of sapphire/Cu joints is calculated using the formula $\tau = F/S$, during the calculation, the area S used is not the entire area of the sapphire, but rather the area of the welded region. Based on the Fig. 10, it can be observed that the width of the weld seam is approximately equal to the gap between the two weld beads. So, we use half of the area of the sapphire as the value of F in our calculations. Under non-optical contact conditions, the shear strength of Joint 1# of sapphire and Cu prepared through direct femtosecond laser-induced micro-welding reaches 46.4 MPa, which ramps up greatly compared with existing welding methods of sapphire and Cu (brazing/diffusion welding). And the shear strength of Joint 2# welded under assistance of Ti coating on sapphire through LIBT grows to 115.5 MPa.

Fig. 10 displays the morphologies of shear fractures in the sapphire side of Joint 1# of sapphire and Cu prepared through direct femtosecond laser-induced micro-welding. It can be seen from Fig. 10(b) and (c) that lots of stripe sapphire that is periodically arranged and shows same welding path is adhered on the fracture surface in the sapphire side, suggesting that the sapphire side near the interface is fractured. Fig. 10 (f) depicts that sapphire on the fracture of Joint 1# exhibits morphologies of brittle intergranular fractures. Plenty of nano-particles are present between two stripes of sapphire on the fracture surface. Composition analysis of these particles reveals that these particles are the mixture of Cu and Al_2O_3 nano-particles. This indicates that there are not only upward splashed nano-particles of ablated Cu, but also downward splashed nano-particles of fused sapphire under the non-optical contact conditions. The droplet morphologies of particles indicate that these particles experience a melting-resolidification process, and they are products of laser ablation.

Fig. 11 shows the morphologies of fractures of Joint 2# of Ti-coated sapphire and Cu prepared through femtosecond laser-induced micro-welding. It can be seen from the figure that a little amount of sapphire is adhered on Cu in the fracture of Joint 2#, this indicates that the joint is mainly fractured at the sapphire/cooper interface, which is probably a result of the reduction of residual stress gradient in the sapphire side near the interface. The element and composition distribution in Fig. 11 shows that Ti also shows periodic stripe distribution on the interface in the fracture. As displayed in Fig. 11(g), Ti on the fracture is clustered in a corner near Al_2O_3 as spherical nano-particles, which indicates that spherical Ti nano-particles tend to enter the molten and mixed zone of Al_2O_3 and Cu in the femtosecond laser-induced micro-welding process.



(f)

Test No.	Element(at.%) Al	O	Cu	Total
1	40.02	56.16	3.81	100
2	29.11	35.64	35.25	100
3	23.76	31.18	45.06	100
4	29.09	37.2	33.72	100
5	37.82	55.07	7.11	100
6	37.31	51.82	10.87	100
7	40.35	54,78	4,87	100
8	13.12	19.04	66.85	100
9	32.64	44.18	23.18	100
10	0	0	100	100

Fig. 6. Locally magnified micro-morphologies and element distribution on cross sections of direct femtosecond laser-induced micro-welded joints. (a), (b) Micro-morphologies in the bonding area; (c) composition distribution in the bonding area; (d) EDS line scanning results; (e) Raman spectra; (f) EDS point scanning results.

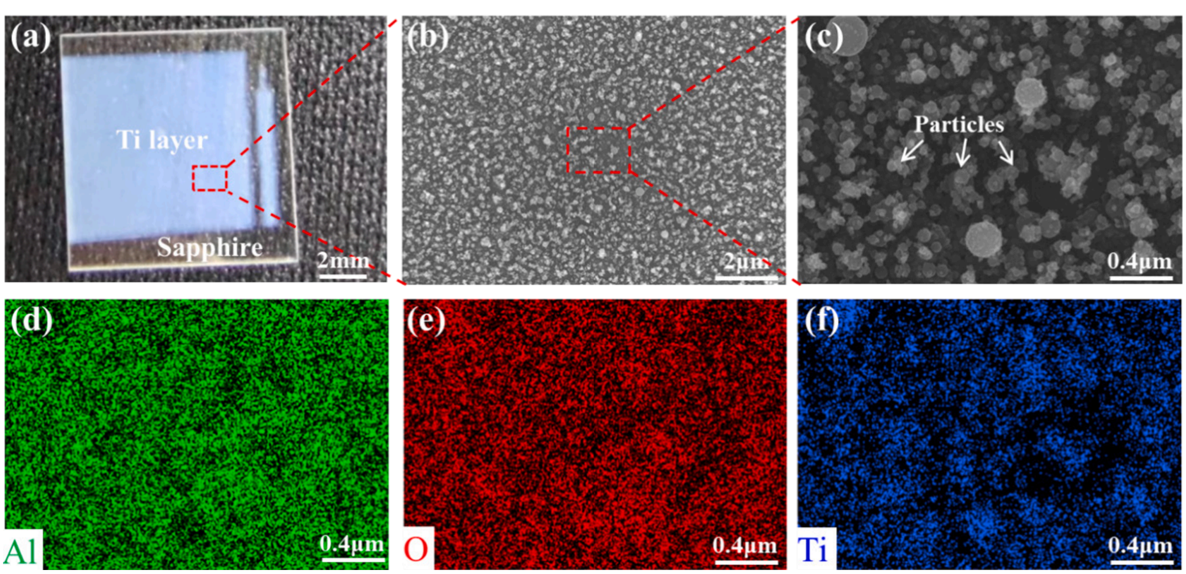


Fig. 7. Ti coating prepared on the sapphire surface through femtosecond LIBT. (a) Surace macro-morphologies; (b)–(c) surface micro-morphologies; (d)–(f) map distribution results of elements.

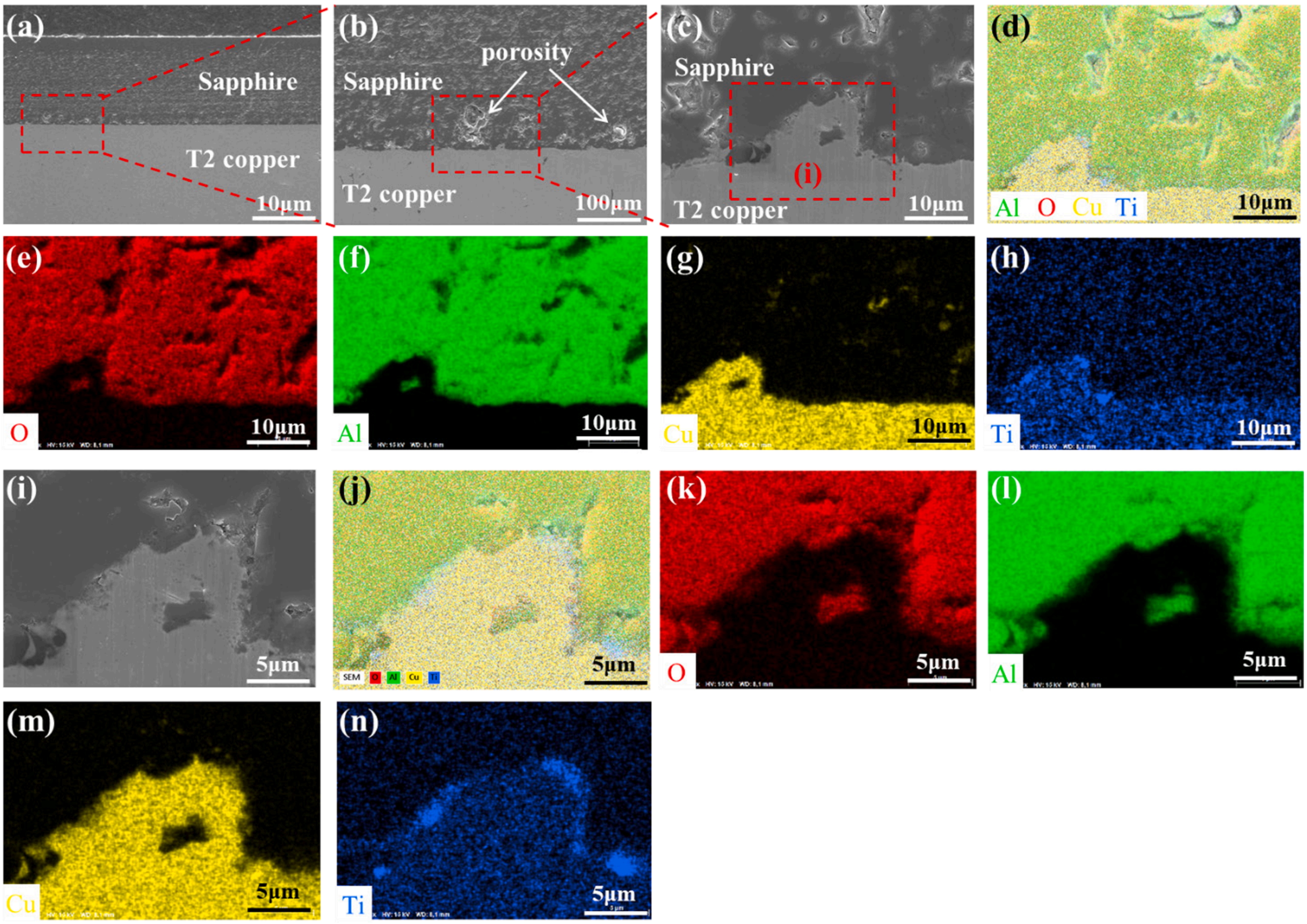


Fig. 8. Cross-sectional morphologies of the joint of Ti-coated sapphire and Cu prepared through femtosecond laser-induced micro-welding. (a)–(c) Micro-morphologies; (d)–(h) element distribution in zone c; (i) typical micro-morphologies; (j)–(n) element distribution in zone i.

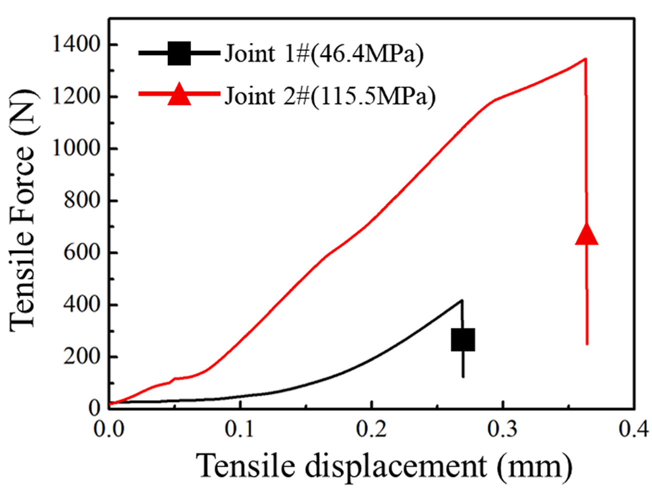


Fig. 9. Shear force–displacement curves of joint 1# (direct femtosecond laser-induced micro-welded joint) and joint 2# (femtosecond laser-induced micro-welded joint of Ti-coated sapphire and Cu).

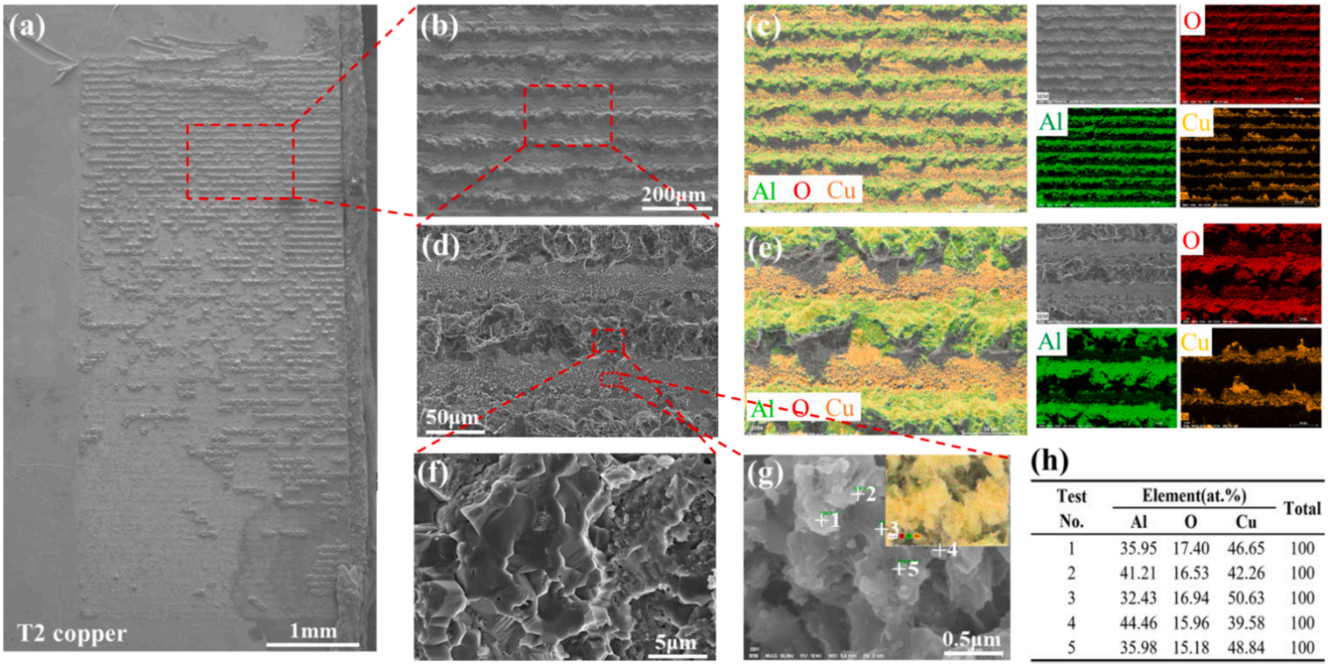
This promotes the bonding of the interface, enables favorable bonding between sapphire and Cu, thus further improving the shear strength of the joint. As displayed in Fig. 11(h) and 11(i), Al_2O_3, Cu, and Ti nano-particles are combined at the interface. EDS point scanning results show that Al_2O_3, Cu, TiO_2, and Ti nano-particles probably exist. This indicates that in the femtosecond laser-induced micro-welding process, the laser-induced fused sapphire, laser-ablated and splashed Cu and Ti, and TiO_2 generated in high-temperature reaction are mixed and bonded at the interface. This improves the bonding force of the bonding interface, effectively relieves residual stress induced by the difference in the coefficient of thermal expansion, and thereby improves the joint strength.

5. Conclusions

In summary, femtosecond laser-induced micro-welding was used to weld sapphire and T2 Cu. Influences of plating Ti coating on sapphire through LIBT on the microstructures and performance of the joints were also investigated. The following main conclusions are obtained:

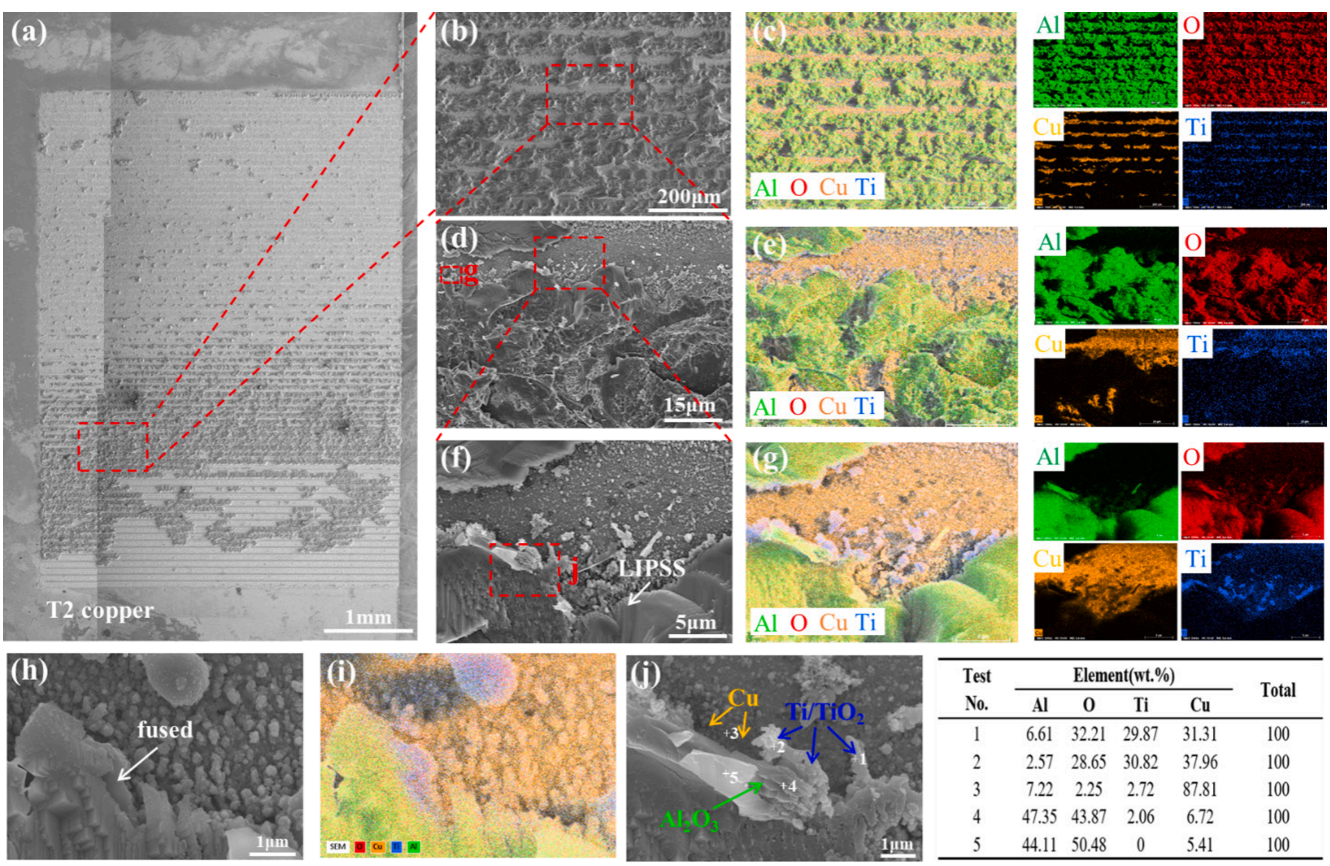
(1) Sapphire and Cu can be directly welded through femtosecond laser-induced micro-welding with the laser power of 15 W, welding speed of 5 mm/s, defocus of 0 mm, and repetition frequency of 500 kHz under non-optical contact conditions. The laser-induced fused sapphire and laser-ablated Cu are mixed at the interface, filling in the gap at the interface and forming effective metallurgical bonding, while no new phases are generated in the sapphire side.

(2) LIBT was adopted to plate Ti coating on the sapphire surface. Ti on the sapphire surface is spherical nanoparticles with the diameter of 20 nm ~ 100 nm and number density of 98 NPs/μm^2. Additionally, well formed micro-welded joints of Ti-coated sapphire and Cu are obtained. A Ti layer of 0.5 μm thick exists at the sapphire/Cu interface, which improves the bonding force



Test No.	Element(at.%) Al	O	Cu	Total
1	35.95	17.40	46.65	100
2	41.21	16.53	42.26	100
3	32.43	16.94	50.63	100
4	44.46	15.96	39.58	100
5	35.98	15.18	48.84	100

Fig. 10. Morphologies of fractures of the direct femtosecond laser-induced micro-welded joint. (a) Overall macro-morphology of the fracture in the sapphire side; (b)–(g) micro-morphologies and compositions on the fractures.



Test No.	Element(wt.%) Al	O	Ti	Cu	Total
1	6.61	32.21	29.87	31.31	100
2	2.57	28.65	30.82	37.96	100
3	7.22	2.25	2.72	87.81	100
4	47.35	43.87	2.06	6.72	100
5	44.11	50.48	0	5.41	100

Fig. 11. Morphologies of fracture of femtosecond laser-induced micro-welded joints of Ti-coated sapphire and Cu. (a) Overall macro-morphology of fracture in the Cu side; (b)–(h) micro-morphologies and composition distribution on fractures.

between sapphire and Cu and is conducive to reducing the residual stress gradient induced by the different coefficients of thermal expansion.

(3) The shear strength of Joint 1# of sapphire and Cu prepared by direct femtosecond laser-induced micro-welding reaches 46.4 MPa, which shows great improvement compared with the

traditional brazing and diffusion welding. The shear strength of Joint 2# welded under assistance of plating Ti coating on the sapphire surface via LIBT further rises to 115.5 MPa.

(4) Periodically distributed sapphire is adhered on the fracture surface in the Cu side of the two types of joints, and the sapphire surface always exhibits morphologies of brittle intergranular fractures. Lots of mixed nano-particles of laser-ablated Cu and Al_2O_3 are present around sapphire. Al_2O_3, Cu. And TiO_2, and Ti nano-particles are combined on the bonding interface on the fracture of the joint 2#. This indicates that in the femtosecond laser-induced micro-welding process, the laser-induced fused sapphire, laser-ablated and splashed Cu and Ti, and TiO_2 produced in high-temperature reactions are mixed and bonded at the interface and improved the bonding force.

CRediT authorship contribution statement

Han Yu: . **Jia-Xuan Zhao:** . **Lin-Jie Zhang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Suck-Joo Na:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

The authors would thank teachers at Instrument Analysis Center of Xi'an Jiaotong University for their assistance with OM, EBSD, SEM and X-ray flaw detection analysis. The authors would thank Mrs. Cheng and Mr. Zhang at Instrument Analysis of Camp Nano.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.optlastec.2024.111063.

References

[1] M.S. Akselrod, F.J. Bruni, Modern trends in crystal growth and new applications of sapphire[J], J. Cryst. Growth 360 (2012) 134–145.
[2] T. Yang, X. Guan, W. Fu, et al., Investigation on symmetric and asymmetric broadband low-loss W-band pillbox windows[J], Electronics 9 (12) (2020) 2060.
[3] P. Hu, W. Lei, Y. Jiang, et al., Development of a 0.32-THz folded waveguide traveling wave tube[J], IEEE Trans. Electron Dev. 65 (6) (2018) 2164–2169.
[4] G.M. Katyba, K.I. Zaytsev, I.N. Dolganova, et al., Sapphire waveguides and fibers for terahertz applications[J], Prog. Cryst. Growth Character. Mater. 67 (3) (2021) 100523.
[5] K.Y. Feng, D.K. Mu, X.J. Liao, et al., Brazing sapphire/sapphire and sapphire/copper sandwich joints using Sn-Ag-Ti active solder alloy[J], Solid State Phenom. 273 (2018) 187–193.
[6] S.V. Gavrish, V.V. Loguinov, S.V. Puchnina, Technology for producing permanent joints between sapphire and metals[J], Weld. Int. 29 (1) (2015) 78–80.
[7] L. He, C. Chen, H. Li, et al., Research advances in joining processes of sapphire[J], Int. J. Adv. Manuf. Technol. 121 (1–2) (2022) 59–81.
[8] P. Li, Y. Yan, J. Ba, et al., The regulation strategy for releasing residual stress in ceramic-metal brazed joints[J], J. Manuf. Process. 85 (2023) 935–947.
[9] T. Tamaki, W. Watanabe, K. Itoh, Laser micro-welding of transparent materials by a localized heat accumulation effect using a femtosecond fiber laser at 1558 nm[J], Opt. Exp. 14 (22) (2006) 10460–10468.
[10] A. Horn, I. Mingareev, J. Gottmann, et al., Dynamical detection of optical phase changes during micro-welding of glass with ultra-short laser radiation[J], Meas. Sci. Technol. 19 (1) (2007) 015302.
[11] A. Horn, I. Mingareev, A. Werth, et al., Investigations on ultrafast welding of glass–glass and glass–silicon[J], Appl. Phys. A 93 (2008) 171–175.
[12] S. Richter, F. Zimmermann, A. Tünnermann, et al., Laser welding of glasses at high repetition rates–Fundamentals and prospects[J], Opt. Laser Technol. 83 (2016) 59–66.
[13] J. Chen, R.M. Carter, R.R. Thomson, et al., Avoiding the requirement for pre-existing optical contact during picosecond laser glass-to-glass welding[J], Opt. Express 23 (14) (2015) 18645–18657.
[14] H. Chen, L. Deng, J. Duan, et al., Picosecond laser welding of glasses with a large gap by a rapid oscillating scan[J], Opt. Lett. 44 (10) (2019) 2570–2573.
[15] X. Jia, K. Li, Z. Li, et al., Multi-scan picosecond laser welding of non-optical contact soda lime glass[J], Opt. Laser Technol. 161 (2023) 109164.
[16] C. Tian, H. Ren, H. Shen, The connection of glass and metal with a large gap by combining laser soldering and ultrafast laser welding[J], J. Manuf. Process. 102 (2023) 528–534.
[17] H. Chen, Y. Rong, Y. Huang, et al., Crack suppression of glass welding by ultrafast laser without optical contact based on light modulation[J], Opt. Laser Technol. 164 (2023) 109466.
[18] L. Zhang, Z. Zhu, T. Xu, et al., Enhanced joint through significant diffusion and molten pool regions in fused silica to aluminum alloy welding by femtosecond mJ-pulses[J], Int. J. Adv. Manuf. Technol. 129 (1) (2023) 601–610.
[19] D. Jayachandran, R. Pendurthi, M.U.K. Sadaf, et al., Three-dimensional integration of two-dimensional field-effect transistors[J], Nature 625 (7994) (2024) 276–281.
[20] J. Zhan, Y. Gao, J. Sun, et al., Mechanism and optimization of femtosecond laser welding fused silica and aluminum[J], Appl. Surf. Sci. 640 (2023) 158327.
[21] R. Pan, D. Yang, T. Zhou, et al., Micro-welding of sapphire and metal by femtosecond laser[J], Ceram. Int. 49 (13) (2023) 21384–21392.
[22] W. Liu, G. Elssner, M. Rühle, Effects of thin film Nb interlayer in Cu/sapphire bonds [J], Mater. Sci. Eng. A 317 (1–2) (2001) 153–162.
[23] R.D. Rodriguez, M. Fatkullin, A. Garcia, et al., Laser-engineered multifunctional graphene-glass electronics[J], Adv. Mater. 34 (43) (2022) 2206877.